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Chapter 1 Introduction

1.1 Project Overview

Life Flow is a Blood Donation Database Management System designed to streamline and organize the information flow between blood donors, recipients, donation campaigns, and blood banks. The system ensures efficient tracking of blood availability, donor eligibility and blood availability.

1.2 Purpose of the Project

The main purpose of Life Flow is to digitize and centralize blood donation records to:

- Improve accessibility of donor and recipient data
- Ensure faster matching of blood groups
- Minimize administrative workload
- Reduce data redundancy and errors
- Support emergency blood request handling

1.3 Scope of the Project

The system covers:

- Recipient registration and tracking
- Donor information management
- Blood group and stock monitoring
- Blood Bank Information with live mapping
- Hospital information management and live mapping
- Record verification and reporting

Chapter 2 System Requirements

2.1 Functional Requirements

The system must:

- Register recipient with complete profile and blood group
- Store recipient details and required blood type
- Manage donation need and match blood
- Record blood stock levels and types
- Generate donor lists and availability of blood in banks

2.2 Non-Functional Requirements

- Security: Only authorized admin access records
- Reliability: Accurate and consistent data handling
- Usability: Simple and intuitive interface
- Scalability: Ability to expand with more records
- Performance: Fast data retrieval

Chapter 3 System Users

- Admin – manages database, updates stock, verifies records
- Recipient – registers, checks eligibility, participates in events
- Donor – information stored from database

Chapter 4 System Design

4.1 UI Design

The following screenshots show the UI design of the system.

4.2 Live Link of the Website

https://singular-cendol-b5cfd3.netlify.app/home

4.3 Simple ER Diagram (Description)

Entities:

- Donor (DonorID, Name, Age, BloodGroup, Contact, LastDonationDate)
- Recipient (RecipientID, Name, BloodGroupNeeded, Contact, UrgencyLevel)
- Donation (DonationID, DonorID, Date, Location)
- Blood Stock (StockID, BloodGroup, Quantity, StorageDate)

Relationships:

- Donor → Donation (1 to Many)
- Donation updates Blood Stock
- Recipient receives blood from Blood Stock

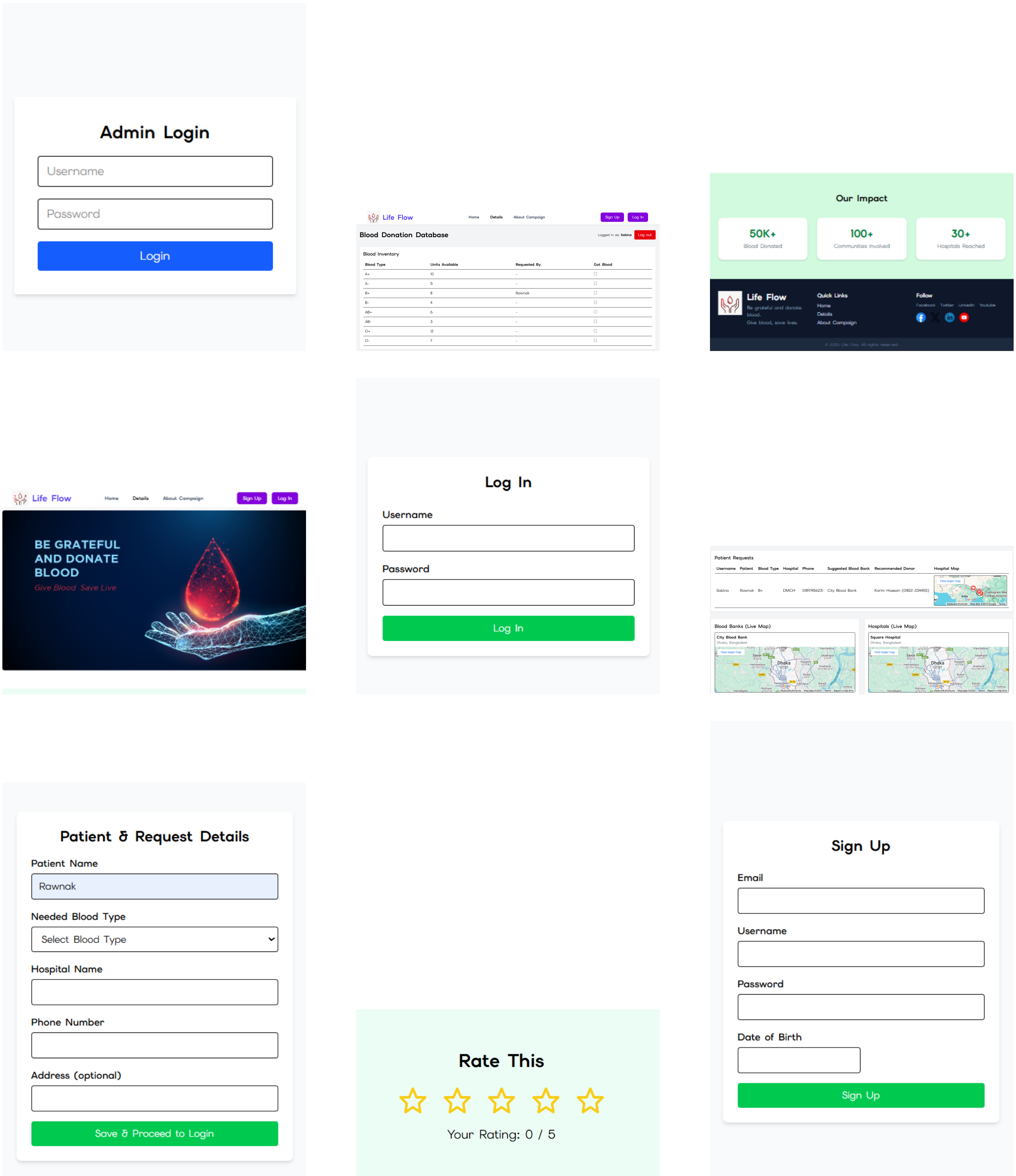

Figure 4.1: UI Design Screenshots (all 9 images)

Chapter 5 Database Structure

5.1 Table Summaries

5.1.1 Donor Table

Stores donor personal and medical information.

5.1.2 Recipient Table

Records individuals requesting blood.

5.1.3 Donation Table

Tracks each donation event and date.

5.1.4 Blood Stock Table

Maintains updated blood inventory levels.

Chapter 6 Data Flow Description

1. Donor registers → System stores profile
2. Admin verifies donor eligibility
3. Donor donates → Donation recorded
4. Blood added to stock
5. Recipient requests blood
6. System matches availability
7. Blood issued and stock updated

Chapter 7 Use Case Descriptions

7.1 Use Case 1: Register Donor

User enters donor details and system stores and confirms registration.

7.2 Use Case 2: Request Blood

Recipient submits request, system checks availability and notifies admin.

7.3 Use Case 3: Update Blood Stock

Admin edits quantity after donation or issue.

Chapter 8 Advantages of the System

- Faster donor–recipient matching
- Organized digital record keeping
- Reduces human error
- Supports emergency response
- Enhances donation campaign logistics

Chapter 9 Framework and Tools Used

9.1 Framework Used

React

9.2 Tools Used

React Router, Toastify, DaisyUI, GSAP, Tailwind, Mapping, Netlify, Github, MySQL Workbench, VS Code, Draw.io

9.3 Language Used

HTML, JavaScript

Chapter 10 Conclusion

Life Flow provides a structured and efficient solution for managing blood donation information. By centralizing data, improving accessibility, and enhancing tracking, it supports life-saving coordination between donors, recipients, and medical staff.